

Project 6

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The training file again, the same 59 returns on 30 stocks used in Projects 1, 2, 4 and 5, with $R_f = 0.001$ carried over from Project 5. All 30 betas are positive, so the filter in (a) removes nothing.

(a)

Stocks ranked by the excess return to beta ratio, with the cut-off values of handout #28:

$$C_i = \frac{\sigma_m^2 \sum_{j=1}^i \frac{(\bar{R}_j - R_f)\beta_j}{\sigma_{\epsilon_j}^2}}{1 + \sigma_m^2 \sum_{j=1}^i \frac{\beta_j^2}{\sigma_{\epsilon_j}^2}}$$

Table 1: Ranking by excess return to beta. ratio = (R-Rf)/beta,
Ai = (R-Rf)beta/s2e, Bi = beta^2/s2e

Stock	R-Rf	beta	ratio	s2e	Ai	sum Ai	Bi	sum Bi	Ci
LLY	0.02521	0.3633	0.06939	0.00500	1.8329	1.8329	26.42	26.42	0.00349
NVO	0.02190	0.4391	0.04987	0.00277	3.4711	5.3040	69.60	96.02	0.00893
NVDA	0.04810	1.3118	0.03667	0.01300	4.8531	10.1571	132.35	228.37	0.01398
NOW	0.03555	1.0001	0.03555	0.00368	9.6706	19.8277	272.05	500.42	0.01985
MSFT	0.02994	0.8625	0.03472	0.00132	19.6002	39.4279	564.56	1064.98	0.02522
NFLX	0.02821	0.8136	0.03468	0.00809	2.8370	42.2649	81.81	1146.79	0.02569
TSLA	0.06819	1.9779	0.03448	0.03163	4.2642	46.5291	123.69	1270.48	0.02631
AMD	0.05680	1.8741	0.03031	0.01921	5.5415	52.0707	182.85	1453.32	0.02668
AAPL	0.03403	1.2019	0.02831	0.00453	9.0317	61.1024	319.00	1772.32	0.02691
AVGO	0.02536	0.9597	0.02642	0.00365	6.6693	67.7717	252.40	2024.72	0.02686
MCD	0.01567	0.5982	0.02620	0.00180	5.1970	72.9687	198.37	2223.09	0.02681
UNH	0.02172	0.8612	0.02522	0.00290	6.4445	79.4133	255.50	2478.60	0.02668
AMZN	0.02606	1.0959	0.02378	0.00413	6.9130	86.3263	290.68	2769.28	0.02642
PFE	0.01604	0.6977	0.02299	0.00400	2.7983	89.1246	121.74	2891.02	0.02630
HD	0.02182	0.9825	0.02221	0.00206	10.4187	99.5434	469.20	3360.22	0.02580
GOOG	0.02314	1.0576	0.02188	0.00200	12.2140	111.7573	558.14	3918.36	0.02530
SBUX	0.01566	0.8235	0.01902	0.00318	4.0618	115.8191	213.58	4131.95	0.02501
V	0.01788	0.9692	0.01845	0.00196	8.8542	124.6733	479.93	4611.88	0.02440

Stock	R-Rf	beta	ratio	s2e	Ai	sum Ai	Bi	sum Bi	Ci
AMGN	0.00977	0.6394	0.01529	0.00371	1.6826	126.3559	110.06	4721.93	0.02421
SCHW	0.01535	1.0448	0.01469	0.00448	3.5817	129.9377	243.82	4965.75	0.02378
META	0.01838	1.2828	0.01433	0.00351	6.7180	136.6557	468.94	5434.69	0.02303
EBAY	0.01579	1.1172	0.01413	0.00514	3.4284	140.0841	242.60	5677.30	0.02268
JNJ	0.00950	0.7387	0.01286	0.00144	4.8684	144.9525	378.52	6055.81	0.02212
JPM	0.01435	1.1351	0.01264	0.00234	6.9682	151.9207	551.09	6606.91	0.02138
MS	0.01886	1.5381	0.01227	0.00247	11.7597	163.6804	958.80	7565.71	0.02030
BAC	0.01566	1.4827	0.01056	0.00239	9.7028	173.3832	918.81	8484.52	0.01930
VZ	0.00453	0.4422	0.01024	0.00153	1.3074	174.6906	127.63	8612.16	0.01917
GS	0.01281	1.4983	0.00855	0.00285	6.7332	181.4238	787.49	9399.65	0.01833
DIS	0.00860	1.1871	0.00724	0.00387	2.6374	184.0612	364.23	9763.88	0.01794
T	-0.00207	0.7345	-0.00281	0.00183	-0.8299	183.2313	294.97	10058.85	0.01736

(b)

With short sales allowed every stock is held and the cut-off is $C^* = C_{30}$. Without short sales, C^* is the last C_i for which the ratio still exceeds it, and only the stocks above that point are held. In both cases

$$z_i = \frac{\beta_i}{\sigma_{\epsilon_i}^2} \left(\frac{\bar{R}_i - R_f}{\beta_i} - C^* \right), \quad x_i = \frac{z_i}{\sum_j z_j}$$

#> short sales allowed : C* = C_30 = 0.017356, 30 stocks, 12 short

#> no short sales : C* = C_9 = 0.026911, 9 stocks held

Table 2: Composition of the point of tangency, in ranked order

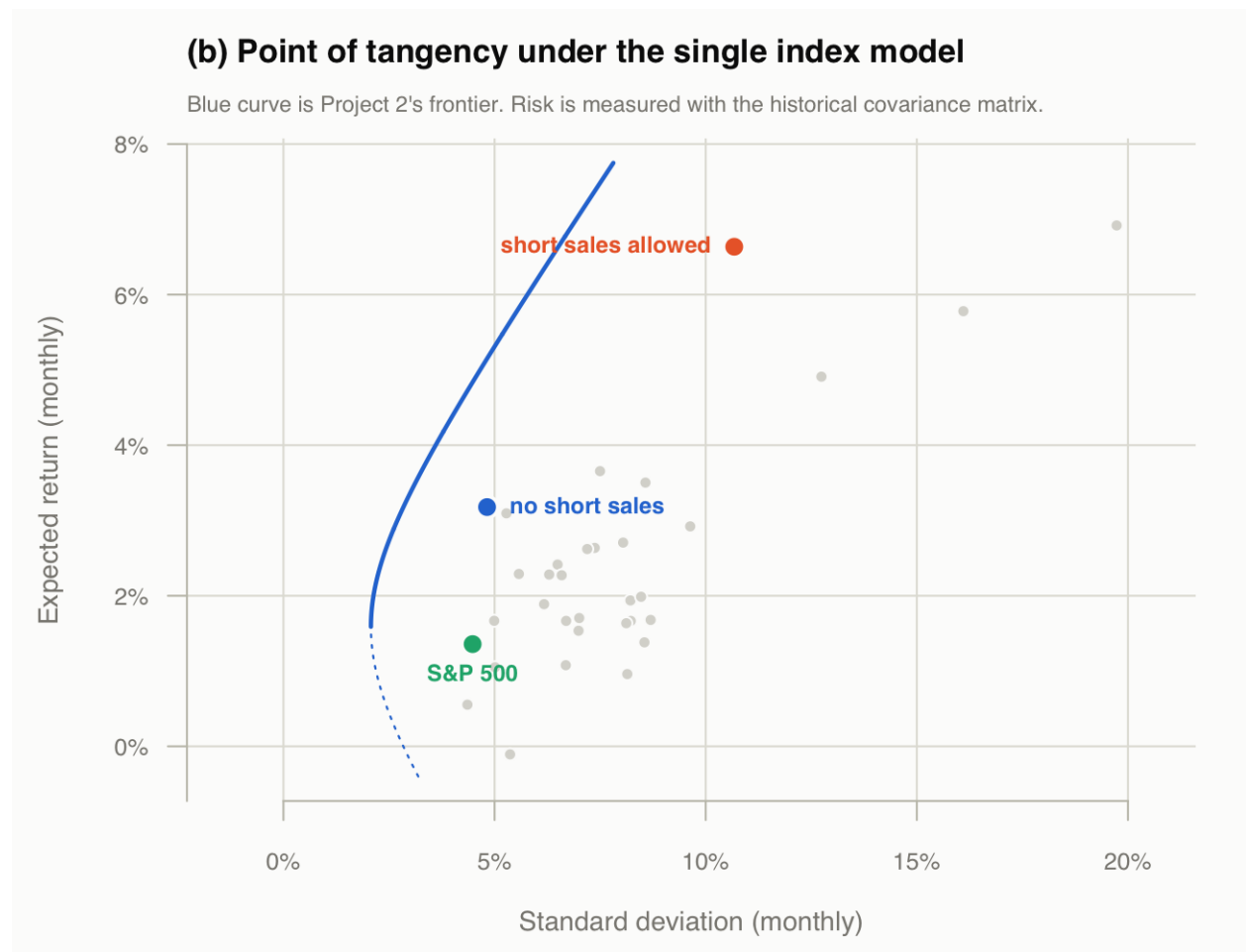
Stock	Short sales	No short sales
LLY	0.2156	0.1803
NVO	0.2937	0.2125
NVDA	0.1110	0.0575
NOW	0.2819	0.1371
MSFT	0.6475	0.2983
NFLX	0.0993	0.0456
TSLA	0.0610	0.0276
AMD	0.0720	0.0193
AAPL	0.1657	0.0217
AVGO	0.1359	0.0000
MCD	0.1671	0.0000
UNH	0.1330	0.0000
AMZN	0.0971	0.0000
PFE	0.0560	0.0000
HD	0.1319	0.0000
GOOG	0.1361	0.0000
SBUX	0.0245	0.0000
V	0.0308	0.0000
AMGN	-0.0203	0.0000
SCHW	-0.0354	0.0000
META	-0.0631	0.0000
EBAY	-0.0399	0.0000
JNJ	-0.1312	0.0000
JPM	-0.1304	0.0000

Stock	Short sales	No short sales
MS	-0.1808	0.0000
BAC	-0.2400	0.0000
VZ	-0.1170	0.0000
GS	-0.2637	0.0000
DIS	-0.1768	0.0000
T	-0.4615	0.0000

```
#>
#> short sales      E    sd (hist)    sd (SIM)
#> no short sales   3.1777%    4.8221%    4.2388%
```

Risk is reported under both matrices: the portfolios are optimised with the single index matrix, but the plots put them alongside the 30 stocks and Project 2's frontier, which are historical.

```
#> max |short sales weights - Project 5 part (a)| = 1.22e-15
```

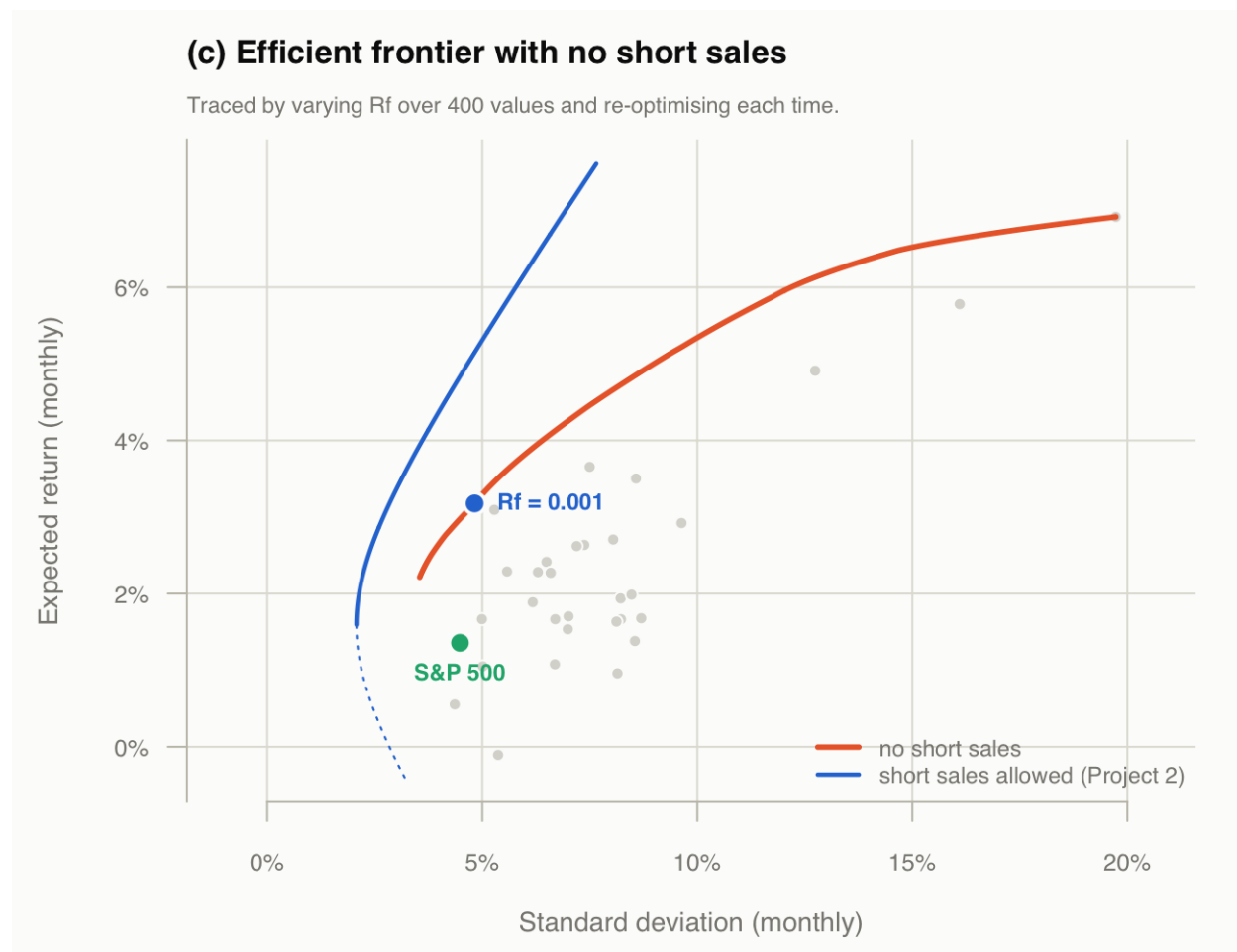


(c)

Varying R_f and re-running the no short sales procedure at each value gives a different tangency portfolio each time; the locus of those points is the efficient frontier under the constraint.

```
#> Rf from -0.0300 to 0.0691, 400 portfolios
```

```
#> E   from 2.215% to 6.919%
#> sd  from 3.544% to 19.732%
#> stocks held: 13 at the lowest Rf, down to 1 at the highest
```



(d)

Under the constant correlation model every pair of stocks shares the same correlation ρ , estimated as the average of the 435 pairwise correlations. Stocks are ranked by the excess return to standard deviation ratio, and

$$C_i = \frac{\rho}{1 - \rho + i\rho} \sum_{j=1}^i \frac{\bar{R}_j - R_f}{\sigma_j}$$

```
#> rho = 0.335146   (pairwise correlations range -0.128 to 0.932)
```

Table 3: Ranking by excess return to standard deviation. ratio = (R-Rf)/sd, multiplier = rho/(1-rho+i*rho)

Stock	R-Rf	sd	ratio	sum ratio	multiplier	Ci
MSFT	0.02994	0.0528	0.56710	0.5671	0.33515	0.19006
NOW	0.03555	0.0750	0.47418	1.0413	0.25102	0.26138

Stock	R-Rf	sd	ratio	sum ratio	multiplier	Ci
AAPL	0.03403	0.0857	0.39693	1.4382	0.20065	0.28858
NVO	0.02190	0.0558	0.39270	1.8309	0.16712	0.30598
NVDA	0.04810	0.1274	0.37756	2.2085	0.14319	0.31623
GOOG	0.02314	0.0649	0.35650	2.5650	0.12525	0.32127
AMD	0.05680	0.1610	0.35274	2.9177	0.11131	0.32477
LLY	0.02521	0.0719	0.35041	3.2681	0.10016	0.32734
HD	0.02182	0.0629	0.34672	3.6148	0.09104	0.32911
TSLA	0.06819	0.1973	0.34557	3.9604	0.08345	0.33048
AVGO	0.02536	0.0737	0.34398	4.3044	0.07702	0.33152
UNH	0.02172	0.0659	0.32967	4.6341	0.07151	0.33139
AMZN	0.02606	0.0804	0.32399	4.9580	0.06674	0.33089
MCD	0.01567	0.0499	0.31400	5.2720	0.06256	0.32984
NFLX	0.02821	0.0963	0.29288	5.5649	0.05888	0.32766
V	0.01788	0.0617	0.28972	5.8546	0.05561	0.32555
SBUX	0.01566	0.0669	0.23394	6.0886	0.05268	0.32073
PFE	0.01604	0.0700	0.22895	6.3175	0.05004	0.31613
META	0.01838	0.0822	0.22365	6.5412	0.04766	0.31173
MS	0.01886	0.0847	0.22274	6.7639	0.04549	0.30768
JPM	0.01435	0.0699	0.20538	6.9693	0.04351	0.30323
BAC	0.01566	0.0822	0.19038	7.1597	0.04169	0.29852
JNJ	0.00950	0.0501	0.18956	7.3492	0.04003	0.29416
SCHW	0.01535	0.0812	0.18906	7.5383	0.03849	0.29012
EBAY	0.01579	0.0870	0.18156	7.7199	0.03706	0.28609
GS	0.01281	0.0855	0.14986	7.8697	0.03573	0.28122
AMGN	0.00977	0.0669	0.14619	8.0159	0.03450	0.27657
DIS	0.00860	0.0814	0.10556	8.1215	0.03335	0.27086
VZ	0.00453	0.0436	0.10397	8.2254	0.03227	0.26548
T	-0.00207	0.0537	-0.03851	8.1869	0.03127	0.25597

(e)

Same cut-off rule, with the constant correlation weights

$$z_i = \frac{1}{(1-\rho)\sigma_i} \left(\frac{\bar{R}_i - R_f}{\sigma_i} - C^* \right)$$

#> short sales allowed : C* = C_30 = 0.255972, 30 stocks, 14 short

#> no short sales : C* = C_11 = 0.331520, 11 stocks held

Table 4: Composition of the point of tangency under constant correlation, in ranked order

Stock	Short sales	No short sales
MSFT	1.2458	0.4531
NOW	0.6154	0.1932
AAPL	0.3476	0.0775
NVO	0.5185	0.1114
NVDA	0.2018	0.0367
GOOG	0.3274	0.0391
AMD	0.1271	0.0134
LLY	0.2775	0.0267

Stock	Short sales	No short sales
HD	0.3049	0.0245
TSLA	0.0960	0.0072
AVGO	0.2524	0.0172
UNH	0.2365	0.0000
AMZN	0.1788	0.0000
MCD	0.2458	0.0000
NFLX	0.0810	0.0000
V	0.1156	0.0000
SBUX	-0.0696	0.0000
PFE	-0.0815	0.0000
META	-0.0832	0.0000
MS	-0.0830	0.0000
JPM	-0.1531	0.0000
BAC	-0.1686	0.0000
JNJ	-0.2801	0.0000
SCHW	-0.1743	0.0000
EBAY	-0.1809	0.0000
GS	-0.2624	0.0000
AMGN	-0.3471	0.0000
DIS	-0.3905	0.0000
VZ	-0.7376	0.0000
T	-1.1600	0.0000

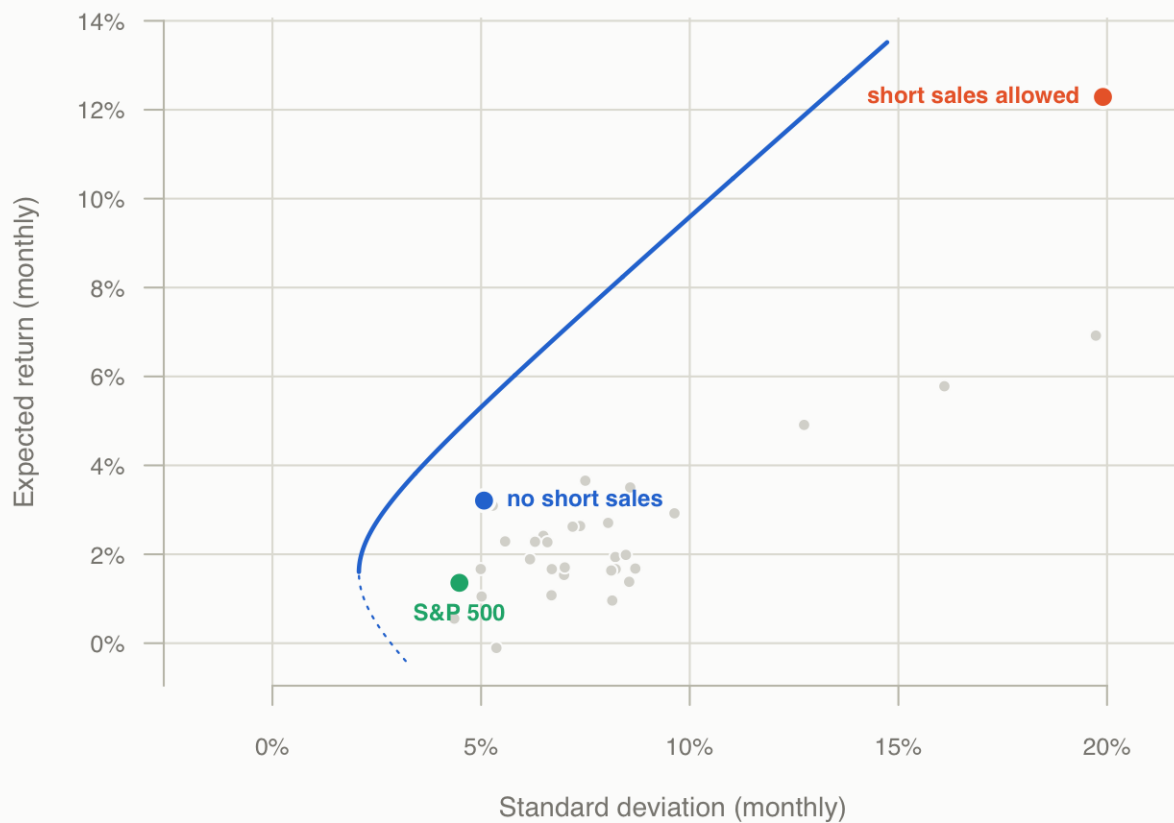
```

#>               E   sd (hist)
#> short sales  12.2879%   19.9015%
#> no short sales  3.2079%    5.0705%

```

(e) Point of tangency under the constant correlation model

Average pairwise correlation $\rho = 0.3351$.



Appendix: full code

```
# Project 6, parts a-e: the Elton-Gruber-Padberg ranking procedure under the
# single index model and under the constant correlation model, with and without
# short sales.

local({
  f <- sub("^--file=", "", grep("^--file=", commandArgs(FALSE), value = TRUE))
  if (length(f) == 1L && basename(f) == "optimization.R")
    setwd(dirname(normalizePath(f)))
})

DATA <- ".."

# ---- data -----
# The training file, the same 59 returns Projects 1, 2, 4 and 5 use.

a <- read.csv(file.path(DATA, "stockData_train.csv"), sep = ",", header = TRUE,
              check.names = FALSE)
stopifnot(!any(duplicated(names(a))))

P <- as.matrix(a[, -1][, -1])
```

```

stopifnot(nrow(P) == 60, ncol(P) == 31, !anyNA(P))

n      <- nrow(P)
ret    <- P[-1, ] / P[-n, ] - 1
stk    <- setdiff(colnames(ret), "^GSPC")
ret30  <- ret[, stk]
Rm     <- as.numeric(ret[, "^GSPC"])
stopifnot(nrow(ret) == 59, length(stk) == 30)

Rbar   <- colMeans(ret30)
sdv    <- apply(ret30, 2, sd)
S_hist <- cov(ret30)

fit    <- lapply(stk, function(s) lm(ret30[, s] ~ Rm))
names(fit) <- stk
beta   <- vapply(fit, function(f) unname(coef(f)[2]), numeric(1))
sig2e  <- vapply(fit, function(f) summary(f)$sigma^2, numeric(1))
sig2m  <- var(Rm)

S_sim  <- outer(beta, beta) * sig2m
diag(S_sim) <- beta^2 * sig2m + sig2e
dimnames(S_sim) <- list(stk, stk)

Rf <- 0.001                                     # the same Rf as Project 5

pos <- beta > 0                                  # "use only the stocks with positive betas"
stopifnot(all(pos))                             # all 30 qualify, so nothing is dropped

# Project 2's frontier, for the plots
frontier <- function(Sig, mu) {
  Si <- solve(Sig); one <- rep(1, length(mu))
  A <- as.numeric(t(one) %*% Si %*% mu); B <- as.numeric(t(mu) %*% Si %*% mu)
  C <- as.numeric(t(one) %*% Si %*% one); D <- B * C - A^2
  list(A = A, B = B, C = C, D = D, Emin = A / C, sdmin = sqrt(1 / C))
}
f_hist <- frontier(S_hist, Rbar)
stopifnot(abs(f_hist$A - 36.9864) < 1e-3, abs(f_hist$C - 2330.03) < 1e-1)
sd_at <- function(f, E) sqrt(1 / f$C + (f$C / f$D) * (E - f$A / f$C)^2)

# round the numeric columns of a table for printing
rnd <- function(d, k = 6) {
  num <- vapply(d, is.numeric, logical(1))
  d[num] <- lapply(d[num], round, k)
  d
}

# portfolio statistics under either covariance matrix
stats <- function(x, Sig) {
  w <- rep(0, 30); names(w) <- stk; w[names(x)] <- x
  c(E = sum(w * Rbar), sd = sqrt(as.numeric(t(w) %*% Sig %*% w)))
}

# ---- (a) ranking by excess return to beta, handout #28 -----

```

```

egp_table <- function(Rf) {
  ratio <- (Rbar - Rf) / beta
  ord <- order(ratio, decreasing = TRUE)
  num <- (Rbar - Rf) * beta / sig2e
  den <- beta^2 / sig2e
  cn <- cumsum(num[ord])
  cd <- cumsum(den[ord])
  data.frame(
    stock = stk[ord],
    Rbar_Rf = (Rbar - Rf)[ord],
    beta = beta[ord],
    ratio = ratio[ord],
    sig2e = sig2e[ord],
    num = num[ord],
    cum_num = cn,
    den = den[ord],
    cum_den = cd,
    Ci = sig2m * cn / (1 + sig2m * cd),
    row.names = NULL
  )
}

tblA <- egp_table(Rf)
stopifnot(!is.unsorted(rev(tblA$ratio))) # ranked high to low

cat("(a) excess return to beta ranking, Rf =", Rf, "\n")
print(rnd(head(tblA[, c("stock", "ratio", "beta", "sig2e", "Ci")], 5)))
cat("    ... \n")
cat(sprintf("    C_n (all 30 stocks) = %.6f\n\n", tblA$Ci[30]))

# ---- (b) tangency portfolio, single index, with and without short sales ----

egp_sim <- function(Rf, short) {
  t0 <- egp_table(Rf)
  if (short) {
    k <- nrow(t0)
  } else {
    ok <- which(t0$ratio > t0$Ci)
    if (!length(ok)) return(NULL)
    k <- max(ok)
  }
  Cstar <- t0$Ci[k]
  inc <- t0$stock[seq_len(k)]
  z <- (beta[inc] / sig2e[inc]) * ((Rbar[inc] - Rf) / beta[inc] - Cstar)
  list(x = z / sum(z), Cstar = Cstar, cutoff = k, table = t0)
}

ss <- egp_sim(Rf, short = TRUE)
noss <- egp_sim(Rf, short = FALSE)

stopifnot(abs(sum(ss$x) - 1) < 1e-12, abs(sum(noss$x) - 1) < 1e-12)
stopifnot(all(noss$x > 0)) # no short positions

```

```

# the short sales answer must reproduce Project 5 part (a), which solved
#  $Z = \Sigma^{-1} (\bar{R} - R_f)$  directly instead of ranking
Z5 <- solve(S_sim, Rbar - Rf)
x5 <- Z5 / sum(Z5)
stopifnot(max(abs(ss$x[stk] - x5[stk])) < 1e-9)

st_ss <- stats(ss$x, S_hist)
st_noss <- stats(noss$x, S_hist)
st_ss_sim <- stats(ss$x, S_sim)
st_noss_sim <- stats(noss$x, S_sim)

cat("(b) point of tangency under the single index model\n")
cat(sprintf("    short sales allowed : C* = %.6f, all %d stocks, %d short\n",
            ss$Cstar, length(ss$x), sum(ss$x < 0)))
cat(sprintf("            E %.4f%% sd %.4f%% (historical) sd %.4f%% (single index)\n",
            st_ss["E"] * 100, st_ss["sd"] * 100, st_ss_sim["sd"] * 100))
cat(sprintf("    short sales not      : C* = %.6f, %d stocks kept of 30\n",
            noss$Cstar, noss$cutoff))
cat(sprintf("            E %.4f%% sd %.4f%% (historical) sd %.4f%% (single index)\n",
            st_noss["E"] * 100, st_noss["sd"] * 100, st_noss_sim["sd"] * 100))
cat("    weights without short sales:\n")
print(round(noss$x, 4))
cat("\n")

# ---- (c) efficient frontier with no short sales, by varying Rf -----

Rf_grid <- seq(-0.03, max(Rbar) - 1e-4, length.out = 400)
noss_fr <- lapply(Rf_grid, function(rf) {
  p <- egp_sim(rf, short = FALSE)
  if (is.null(p)) return(NULL)
  s <- stats(p$x, S_hist)
  c(Rf = rf, E = s["E"], sd = s["sd"], k = p$cutoff)
})
noss_fr <- as.data.frame(do.call(rbind, noss_fr[!vapply(noss_fr, is.null, logical(1))]))
names(noss_fr) <- c("Rf", "E", "sd", "k")

# every one of these portfolios is long only and fully invested, and none can
# sit inside the unconstrained frontier
stopifnot(all(noss_fr$sd >= sd_at(f_hist, noss_fr$E) - 1e-9))

cat("(c) no short sales frontier traced over", nrow(noss_fr), "values of Rf\n")
cat(sprintf("    Rf from %.4f to %.4f\n", min(noss_fr$Rf), max(noss_fr$Rf)))
cat(sprintf("    E  from %.4f%% to %.4f%%\n", min(noss_fr$E) * 100, max(noss_fr$E) * 100))
cat(sprintf("    stocks held: %d down to %d\n", max(noss_fr$k), min(noss_fr$k)))

# ---- (d) constant correlation model, handout #33 -----

CC <- cor(ret30)
rho <- mean(CC[upper.tri(CC)])

cc_table <- function(Rf) {
  ratio <- (Rbar - Rf) / sdv
  ord <- order(ratio, decreasing = TRUE)

```

```

i      <- seq_along(ord)
cum    <- cumsum(ratio[ord])
data.frame(
  stock      = stk[ord],
  Rbar_Rf    = (Rbar - Rf)[ord],
  sd         = sdv[ord],
  ratio      = ratio[ord],
  cum_ratio  = cum,
  mult       = rho / (1 - rho + i * rho),
  Ci         = (rho / (1 - rho + i * rho)) * cum,
  row.names = NULL
)
}

tblD <- cc_table(Rf)
stopifnot(!is.unsorted(rev(tblD$ratio)))

cat("(d) constant correlation model, Rf =", Rf, "\n")
cat(sprintf("    average pairwise correlation rho = %.6f\n", rho))
print(rnd(head(tblD[, c("stock", "ratio", "cum_ratio", "mult", "Ci")], 5)))
cat("    ... \n")
cat(sprintf("    C_n (all 30 stocks) = %.6f\n", tblD$Ci[30]))

# ---- (e) tangency portfolio, constant correlation -----

egp_cc <- function(Rf, short) {
  t0 <- cc_table(Rf)
  if (short) {
    k <- nrow(t0)
  } else {
    ok <- which(t0$ratio > t0$Ci)
    if (!length(ok)) return(NULL)
    k <- max(ok)
  }
  Cstar <- t0$Ci[k]
  inc  <- t0$stock[seq_len(k)]
  z    <- (1 / ((1 - rho) * sdv[inc])) * ((Rbar[inc] - Rf) / sdv[inc] - Cstar)
  list(x = z / sum(z), Cstar = Cstar, cutoff = k, table = t0)
}

cc_ss <- egp_cc(Rf, short = TRUE)
cc_noss <- egp_cc(Rf, short = FALSE)

stopifnot(abs(sum(cc_ss$x) - 1) < 1e-12, abs(sum(cc_noss$x) - 1) < 1e-12)
stopifnot(all(cc_noss$x > 0))

# the constant correlation matrix implied by rho, as a cross-check on the
# short sales solution: it must equal Sigma_cc^-1 (Rbar - Rf 1) rescaled
S_cc <- rho * outer(sdv, sdv)
diag(S_cc) <- sdv^2
Zcc <- solve(S_cc, Rbar - Rf)
stopifnot(max(abs(cc_ss$x[stk] - (Zcc / sum(Zcc))[stk])) < 1e-9)

```

```

st_cc_ss <- stats(cc_ss$x, S_hist)
st_cc_noss <- stats(cc_noss$x, S_hist)

cat("(e) point of tangency under the constant correlation model\n")
cat(sprintf("    short sales allowed : C* = %.6f, all %d stocks, %d short\n",
            cc_ss$Cstar, length(cc_ss$x), sum(cc_ss$x < 0)))
cat(sprintf("                E %.4f%%    sd %.4f%%\n",
            st_cc_ss["E"] * 100, st_cc_ss["sd"] * 100))
cat(sprintf("    short sales not      : C* = %.6f, %d stocks kept of 30\n",
            cc_noss$Cstar, cc_noss$cutoff))
cat(sprintf("                E %.4f%%    sd %.4f%%\n",
            st_cc_noss["E"] * 100, st_cc_noss["sd"] * 100))
cat("    weights without short sales:\n")
print(round(cc_noss$x, 4))
cat("\n")
# ---- plotting helpers -----

surface <- "#fcfcfb"; ink <- "#0b0b0b"; muted <- "#898781"; grid <- "#e1e0d9"
c_front <- "#2a78d6"; c_eq <- "#eb6834"; c_spx <- "#1baf7a"; c_arb <- "#6f6d67"

open_png <- function(f, h = 4.7) {
  png(f, width = 6.5, height = h, units = "in", res = 200)
  par(bg = surface, mar = c(4.2, 4.6, 3.4, 1.4), family = "sans")
}
ax <- function(side, at, lab) {
  axis(side, at = at, labels = lab, col = "#c3c2b7", col.axis = muted,
       cex.axis = 0.75, lwd = 1, las = if (side == 2) 1 else 0)
}
pt <- function(x, y, col, lab, pos = 4, cex = 0.72, off = 0.6) {
  points(x, y, pch = 21, bg = col, col = surface, lwd = 1.8, cex = 1.5)
  text(x, y, lab, pos = pos, offset = off, cex = cex, font = 2, col = col)
}

# the common backdrop: 30 stocks, the index, and Project 2's frontier
backdrop <- function(xr, yr, xtick) {
  plot(NA, xlim = xr, ylim = yr, axes = FALSE, xlab = "", ylab = "")
  abline(h = pretty(yr), v = xtick, col = grid, lwd = 1)
  Eg <- seq(yr[1], yr[2], length.out = 800)
  sdg <- sd_at(f_hist, Eg)
  eff <- Eg >= f_hist$Emin
  lines(sdg[!eff], Eg[!eff], col = c_front, lwd = 1.1, lty = 3)
  lines(sdg[ eff], Eg[ eff], col = c_front, lwd = 2.2)
  points(sdv, Rbar, pch = 21, bg = "#d9d8d2", col = surface, lwd = 1, cex = 0.85)
  points(sd(Rm), mean(Rm), pch = 21, bg = c_spx, col = surface, lwd = 1.8, cex = 1.5)
  text(sd(Rm), mean(Rm), "S&P 500", pos = 1, offset = 0.6, cex = 0.72,
       font = 2, col = c_spx)
  ax(1, xtick, sprintf("%.0f%%", xtick * 100))
  ax(2, pretty(yr), sprintf("%.0f%%", pretty(yr) * 100))
  mtext("Standard deviation (monthly)", 1, line = 2.6, col = muted, cex = 0.85)
  mtext("Expected return (monthly)", 2, line = 3.2, col = muted, cex = 0.85)
}

# ---- (b) single index tangency portfolios -----

```

```

open_png("fig_sim_tangency.png", h = 5.0)
xr <- c(-0.014, max(c(sdv, st_ss["sd"]))) * 1.05)
yr <- c(-0.004, max(c(Rbar, st_ss["E"]))) * 1.12)
backdrop(xr, yr, pretty(c(0, max(sdv))))
pt(st_ss["sd"], st_ss["E"], c_eq, "short sales allowed", 2)
pt(st_noss["sd"], st_noss["E"], c_front, "no short sales", 4)
mtext("(b) Point of tangency under the single index model", 3, line = 1.8,
      adj = 0, col = ink, cex = 1.02, font = 2)
mtext("Blue curve is Project 2's frontier. Risk is measured with the historical covariance matrix.",
      3, line = 0.5, adj = 0, col = muted, cex = 0.7)
dev.off()

# ---- (c) the no short sales frontier -----

open_png("fig_noshort_frontier.png", h = 5.0)
o <- order(noss_fr$E)
xr <- c(-0.010, max(sdv) * 1.05)
yr <- c(-0.004, max(c(Rbar, noss_fr$E)) * 1.10)
backdrop(xr, yr, pretty(c(0, max(sdv))))
lines(noss_fr$sd[o], noss_fr$E[o], col = c_eq, lwd = 2.8)
pt(st_noss["sd"], st_noss["E"], c_front, sprintf("Rf = %.3f", Rf), 4)
legend("bottomright", bty = "n", cex = 0.72, text.col = muted,
      lwd = c(2.8, 2.2), col = c(c_eq, c_front),
      legend = c("no short sales", "short sales allowed (Project 2)"))
mtext("(c) Efficient frontier with no short sales", 3, line = 1.8, adj = 0,
      col = ink, cex = 1.02, font = 2)
mtext(sprintf("Traced by varying Rf over %d values and re-optimising each time.",
              nrow(noss_fr)), 3, line = 0.5, adj = 0, col = muted, cex = 0.7)
dev.off()

# ---- (e) constant correlation tangency portfolios -----

open_png("fig_cc_tangency.png", h = 5.0)
xr <- c(-0.017, max(c(sdv, st_cc_ss["sd"]))) * 1.05)
yr <- c(-0.004, max(c(Rbar, st_cc_ss["E"]))) * 1.10)
backdrop(xr, yr, pretty(c(0, max(c(sdv, st_cc_ss["sd"]))))))
pt(st_cc_ss["sd"], st_cc_ss["E"], c_eq, "short sales allowed", 2)
pt(st_cc_noss["sd"], st_cc_noss["E"], c_front, "no short sales", 4)
mtext("(e) Point of tangency under the constant correlation model", 3, line = 1.8,
      adj = 0, col = ink, cex = 1.02, font = 2)
mtext(sprintf("Average pairwise correlation rho = %.4f.", rho),
      3, line = 0.5, adj = 0, col = muted, cex = 0.7)
dev.off()

cat("wrote fig_sim_tangency.png, fig_noshort_frontier.png, fig_cc_tangency.png\n")

```